

A Survey on Network Tomography Applications

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Abstract— Network tomography is a technique that uses end-points values to determine the inside state of a network. In this paper, we survey the fields in which network tomography has been applied, including computer networking, traffic monitoring, medical imaging, and communications. We discuss the applications of network tomography in each field and highlight some notable research efforts in the area. Network tomography has been used in computer networking to infer individual link performance metrics, network topology, and end-points values. In traffic screening, it has been used to optimize urban planning and traffic flow. In medical imaging, it has been used to improve image reconstruction. In communications, it has been used to enhance wireless network performance.

Keywords— Network tomography, Computer networking, Network performance.

I. INTRODUCTION

Network tomography is a technique that uses end-points values to determine the inside situation of a network. In this survey, we cover how network tomography is used in several fields. This technique has been implemented in a many fields, including computer networking, traffic monitoring, medical imaging, and communications.

In the field of computer networking, network tomography has been used to infer individual link performance metrics, network topology [1], and end-points values from incomplete sets of data. For choosing paths in network tomography when there are network element failures and varied probing costs, researchers have suggested optimization algorithms [2].

Network tomography has been applied in the field of traffic monitoring to improve the live in urbans. This is done by using network tomography to optimize traffic flow, road construction, and urban planning [3]. Traffic monitoring has previously been done using sensors, probe vehicles, and crowd sensing technologies, but these methods can be costly, limited in scope, and potentially harmful to user privacy [3]. Network tomography offers an alternative approach by using end-points values to estimate the inside situation of a network [4][5].

In the field of medical imaging, network tomography has been used to reconstruct images of internal organs and tissues based on data from medical scanners. By combining measurements from multiple sources, network tomography can produce high-resolution images that provide valuable information for medical diagnosis and treatment [6], [7].

In the field of communications, network tomography has been used to enhance wireless network performance by inferring link performance metrics and network topology [8]; evaluating internal quality of service (QoS) attributes, traffic

volumes, or the network only requires the observation of a small section of network nodes [9], [11].

In this survey, we give an overview of the various fields in which network tomography has been applied and discuss the applications of network tomography in each field. We also highlight some notable research efforts in the area of network tomography.

II. LITERATURE OVERVIEW

This section is divided into subsections according to the field in which network tomography is used. We begin by discussing the useage of network tomography in the field of computer networking. Then, in the field of traffic monitoring. After that, we review the useage of network tomography in medical imaging. nally, we discuss the useage of network tomography in network communications. Overall, this literature review provides an overview of the recent work that has adopted network tomography in different fields and discusses the potential of this tool for future applications. We hope to provide a thorough overview of the status of the field by organizing this work and highlighting the potential of network tomography for further research and development.

A. Network Tomography in Computer Networking

Network tomography has been used to improve network performance by estimating network parameters such as link capacities and delays. This information can be used by network administrators to optimize routing and congestion control algorithms, leading to better network performance.

In their study, the authors of [2] address the issue of selecting paths in network tomography to make the performance better even with network's failures. Network tomography allows for the acquisition of information about a network without the need for direct monitoring of individual network elements. This is done by using probes at the edges of the network to measure end-points paths, which enables the efficient interpretation of single link performance metrics, network topology interpretation, and estimation of the values of the entire collection of endpoints from a partial set. However, network element failures are a common occurrence and can negatively impact the performance in several network tomography applications. To address this issue, the authors propose an optimization problem that takes into account both robustness and probing cost make better performance in

network tomography operations, even with the presents of failures. This will improve the network tomography works during the select of the paths. They aim to maximize the expected rank (a metric used to model the paths' robustness in network tomography) under a constraint budget of the probing cost. The issue is NP-Hard, but the authors suggested the use of RoMe algorithm with high approximation ratio to solve it. They also provide a limit that can be efficiently evaluated for cases where evaluating the expected rank is difficult. In addition, the authors look at the case where the error distribution is unknown and suggest an algorithm that based on reinforcement learning for the optimization problem, with the use of RoMe algorithm as a second procedure. They compare their solution to the most advanced path selection algorithm currently available, showing how their techniques improve the performance, significantly, for the applications in network tomography with the existence of errors. Overall, despite the failures in network and varied costs, the authors' work offers a novel method for improving path selection in network tomography.

The method presented in [1], presents a new method for inferring the internal network's parameters, such as delays in link and dropping of packet rates, using network tomography. Each research packet is cloned and delivered down each network branch path during multicast transmission, which is the paper's main area of interest. The correlation in performance noticed on overlapping end-point paths can be used to the multicast values' advantage. The authors in [1], propose a new method for inferring network delay performance using the method of moments. This method is computationally simple, but its accuracy is not very high. To improve the accuracy of the method, the authors suggest a new method that uses a hybrid approach combining the method of moments with a Maximum Likelihood Estimator (MLE). This new method is able to improve the accuracy of delay distribution inference while still maintaining a low computational complexity, as shown in Figure 1 from [1]. The authors assess the performance of the presented solution using simulations. The results demonstrate that the suggested method is more accurate and computationally efficient than the existing methods.

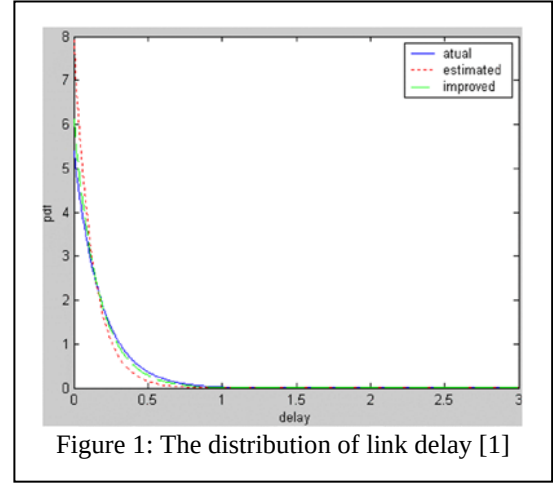


Figure 1: The distribution of link delay [1]

B. Network Tomography in Traffic Monitoring

Network tomography has been used to estimate traffic flow and congestion levels on roads and highways. This information can be used by traffic managers to optimize traffic patterns and reduce congestion, improving the efficiency of the transportation network.

In [4], a method for effective network tomography-based urban traffic monitoring is presented through an analysis of the correlation between connection latency and the trip times of road segments. The system employs edge intelligence to optimize data collection and processing in order to provide more accurate and knowledge of the time of urban travel time. This is achieved by tackling two problems related to finite telecommunications sources: (1) when all of videos that can be processed into travel time are defined, a Fisher Information Matrix will be used to create the best strategy for each traffic detector, and (2) when centralized processing alone is insufficient, computation capabilities from traffic detectors are utilized handling clever computation offloading by use of a multi-agent reinforcement learning method. Numerous simulations show that the suggested strategy, when compared to well-known alternatives using either uniform or random strategies, effectively lowers estimation error rate.

The authors in [3], present an innovative method of monitoring traffic in smart cities that relies on a finite number of cameras located at intersections of the roads to calculate car's start and end points of traveling times. This approach does not require probe vehicles or GPS localization, which are most of the time not efficient in terms of cost, flexibility, and scalability. Instead, it uses the analytical scheme of network tomography to figure out the time for travelling of all sections of the road in the network, taking into account potential noise in the assessments and the unexpected paths of vehicles. The best camera placement for maximizing cover with minimizing errors and the cost is another topic covered in the study. The topography of downtown of the city of San Francisco is extensively experimentally evaluated using real data that gathered through Google Maps and using synthetic networks. The findings demonstrate that, despite just 20% of

road crossings having cameras, the suggested approach can accurately estimate trip times over 95% of all roadways [3]. This is a significant improvement over traditional techniques, which are often expensive and not scalable.

C. Network Tomography in Medical Imaging

Network tomography is used in medical imaging to create and reconstruct detailed images of internal organs and tissues. The technique uses data from medical scanners and combines measurements from multiple sources to produce high-resolution images that can be used for medical diagnosis and treatment.

The study in [6] suggests a self-training method for bioluminescence tomography (BLT) reconstruction, a non-invasive imaging method used in pre-clinical research to detect physiological activity in the bodies of living beings. The proposed approach rapidly generates large data sets with unpredictable objective sizes, shapes, and numbers using a growth of seed mechanism randomly, enabling the neural network to self-train and become more adaptive in its capacity to recreate forms out-side of the training data. Using this method shows an improvement on the performance of BLT reconstruction. The use of self-training in this context is related to the broader topic of network tomography. It includes estimating a network's internal state via network measurements. In BLT, the neural network is taught to infer, from measurements of photon intensity on the body's surface, the distribution of bioluminescent sources within the body. In Figure 2, which is shown in [6], the proposed network's loss function is contrasted with the end-points network's loss function as a function of the epoch number, where stands for the learning rate.

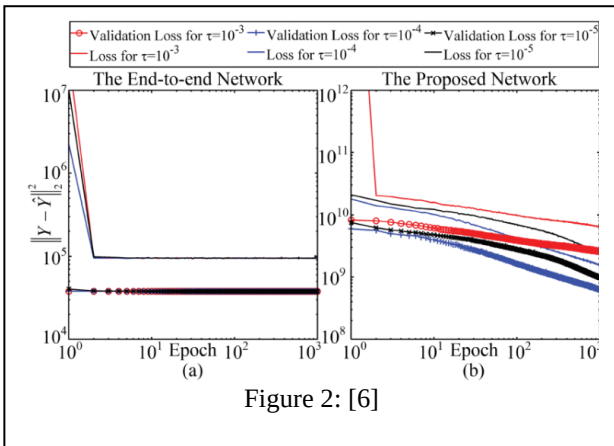


Figure 2: [6]

D. Network Tomography in Communications

Network tomography has been used for diagnosing and troubleshooting network problems. By estimating the internal state of a network from externally observable patterns of communication, network tomography can provide valuable insights into network performance. This can help communication engineers identify bottlenecks and other

performance issues, allowing them to design more efficient and robust communication protocols. Network tomography can also be used for real-time monitoring and diagnostic purposes, making it a valuable tool for improving network performance and reliability.

The authors of [9], discuss how network tomography can be used in combination with software-defined networking (SDN) to improve the monitoring of 5G networks. The authors argue that network tomography can complement and work together with SDN to yield accurate performance estimations while exploiting the capabilities of SDN, like the centralized perspective of the network and the ability to make direct flow-level measurements. The authors present a range of applications for network tomography-based solutions in 5G networks, including virtual networks and vehicular networks. They also discuss how network tomography can be used to address challenges in 5G networks, such as the need for low-overhead monitoring and the dynamic nature of network environments. The authors demonstrate that network tomography can be part of computing key performance indicators for vertical applications running on 5G networks, providing information that can be used for engineering the traffic, error detection and management, balancing of the load, and service level agreement compliance. They conclude that network tomography can provide significant benefits for monitoring 5G networks and can be used in combination with SDN to provide accurate performance estimations with low overhead. They suggest that the application of network tomography-based solutions could be advantageous in a range of different network scenarios, including virtual, data center, car, and cloud networks.

A new network topology inference method that combines the benefits of tomography and tracerouting is presented in the study in [8]. When estimating network performance and building networks, network topology inference is used to infer the underlying structure of a network. Traditional techniques of network topology inference utilizing traceroute rely on router collaboration, but due to the internet's explosive growth and heightened security needs, anonymous routers that do not collaborate have emerged. The accuracy of network topology inference using traceroute may be significantly impacted by this. Tomography-based network topology inference, in contrast, does not rely on router cooperation and only requires measurements from two endpoints. This technique, which determines the network architecture through active probing or passive data gathering, has gained popularity in recent years. However, it has several drawbacks, including the need to send a large number of probe packets to obtain statistical information about links, the use of complex statistical algorithms that can be time-consuming, and the uncertainty of the results due to statistical laws [8]. The new method introduced in the paper addresses these problems by combining the advantages of tracerouting and tomography. A preliminary topology is first created using traceroute results, and a final topology is then obtained using a "topology updating algorithm" and "sandwich" probes. The approach is demonstrated to be successful in addressing the issue of anonymous routers and enhancing the precision of network topology inference. It also reduces the number of packets that need to be sent and the time required for calculations [8]. The results of simulations on Network Simulator (NS2) show that

the method has good performance and can be applied to real-world networks.

The paper in [11], proposes an efficient network path loss rate inference method that requires fewer measurements than existing systems and has higher accuracy. This is important because accurate and efficient measurement of network performance is crucial for ensuring the quality of network monitoring and reducing management costs. Existing methods for measuring network performance require $O(n \log n)$ measurements [11], where n is equal to hosts' number in the network. This can be time-consuming and costly. The proposed method aims to reduce the number of measurements needed by inferring the loss rate of all remaining paths with higher accuracy using only a subset of paths with uniform distribution. The new method uses network tomography for inferring the inside condition of a network from externally observable patterns of communication. This allows the method to adaptively select the next path to measure using the results of the previous probe, reducing the complexity of the algorithm. The proposed method also simplifies the matrix of routing information by removing non-congested paths and links, reducing the number of independent paths needed for inference. This makes the algorithm more efficient and reduces the complexity of the calculation. Simulation results, as shown in Figure 3 and Figure 4, show that the proposed method can save 5% to 14% of probing paths compared to existing methods. Additionally, the new method can obtain the loss rates of all remaining paths without solving for the value of the link loss rate, further reducing the complexity of the algorithm [11]. Overall, the suggested method is a useful tool for network performance monitoring since it can effectively predict network path loss rates with fewer measurements and at a reduced cost.

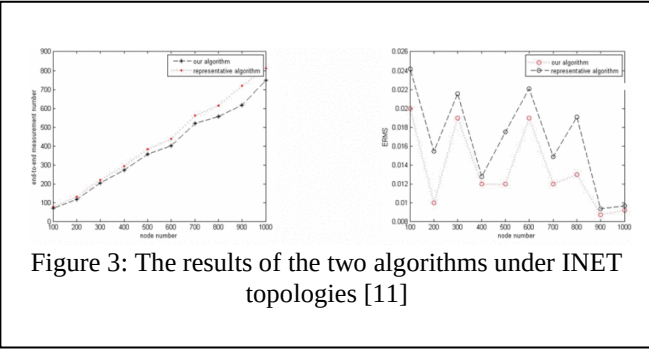


Figure 3: The results of the two algorithms under INET topologies [11]

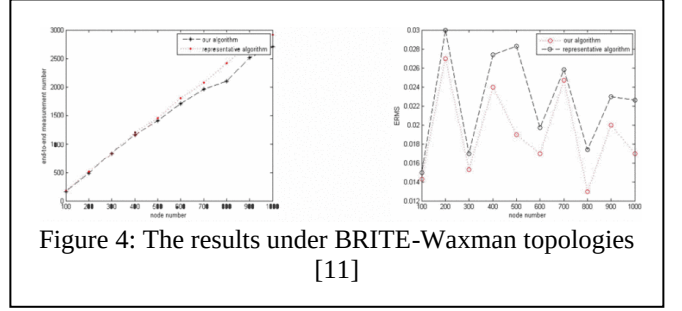


Figure 4: The results under BRITE-Waxman topologies [11]

III. CRITICAL ANALYSIS

One limitation of current applications in network tomography is that they often require many probes or values to accurately infer the inside condition of a network. This can be costly and time-consuming, and may not be practical in some scenarios. For example, in the field of computer networking, network tomography has been used to infer individual link performance metrics and network topology [1], but these approaches often require many probes or measurements to achieve accurate results. This can be a barrier to the wider adoption of network tomography, as the cost and time required for these measurements may not be feasible in some cases.

Additionally, current approaches may not be able to handle large-scale networks or networks with complex topologies. This restricts the usage of network tomography to only specific network architectures and use cases. For example, network tomography may be impractical for networks with a large number of nodes or links, or for networks with complex topologies that cannot be accurately modeled using current methods. This can prevent communication engineers from using network tomography to diagnose and troubleshoot network problems in these scenarios, limiting the potential benefits of this powerful tool.

One potential future direction for network tomography research is to develop more efficient algorithms that require fewer probes or values to infer the inside situation of a network. This could potentially reduce the cost and time required for network tomography, making it more practical to apply in a wider range of scenarios. Additionally, this could enable the use of network tomography in scenarios where it is currently impractical due to the large number of probes or measurements required. For example, efficient algorithms could enable the use of network tomography in large-scale networks or networks with complex topologies, allowing communication engineers to diagnose and troubleshoot network problems in these scenarios.

Another potential future direction is to develop approaches that can handle large-scale networks or networks with complex topologies. This could allow network tomography to be applied to a wider range of networks and enable the inference of network states in scenarios where current approaches are not effective. For example, new algorithms could be developed that are able to accurately model and infer the internal state of large-scale or complex networks, allowing communication engineers to use network tomography to diagnose and troubleshoot network problems in these scenarios. This would expand the range of applications for

network tomography, making it a more powerful and versatile tool for communication engineers.

IV. CONCLUSION

In conclusion, network tomography is a great method to understand the inside condition of a network from outside observable patterns of communication. This approach has been utilized in a range of fields, including computer networking, traffic monitoring, medical imaging, and communications, with the goal of improving network performance and optimizing traffic flow. In the field of computer networking, network tomography has been used to infer individual link performance metrics and network topology, enabling the design of more efficient and robust communication protocols. In traffic monitoring, network tomography has been used to infer the state of transportation networks, allowing transportation planners to design more efficient and effective transportation systems. In medical imaging, network tomography has been used to help reconstructing medical images that can be helpful for treating and diagnosing disease and cancer. And in communications, network tomography has been used to improve wireless network performance, enhancing the reliability and robustness of wireless networks.

This survey paper provides an overview of the various fields in which network tomography has been applied and highlights some notable research efforts in the area. We have presented a review of the recent work that has adopted network tomography in different fields, discussing the potential of this tool for future applications. We believe that network tomography has great potential to continue to improve network performance and optimize traffic flow, and we look forward to future research and development in this exciting field.

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